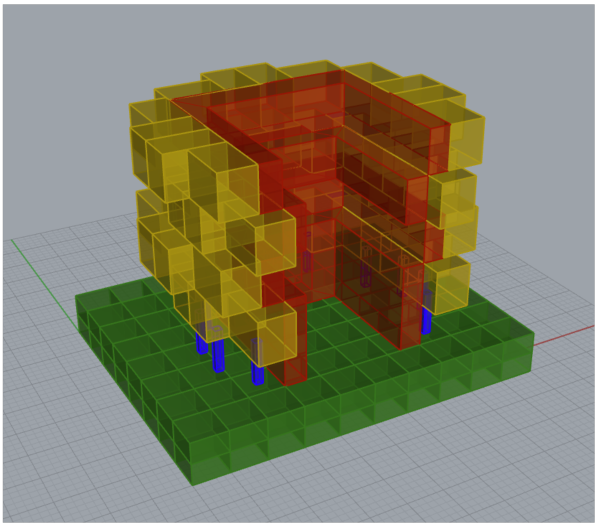
# Homework 03

Building Graph Representation

*Graph Machine Learning — Session 06 (13A & 13B)*



*Building overview — CellComplex coloured by spatial type*

## The Building Model

The building was modelled in Rhinoceros 3D and exported as four separate OBJ files, one for each spatial category: ground slab, structural columns, office volumes, and service core. Each file corresponds to a distinct node type in the graph and is loaded independently in the notebook.

## Part 1 — Constructing the Graph (13A)

### Loading and Tagging Geometry

Each OBJ file was imported using `Topology.ByOBJPath` with `transposeAxes=True` to convert from OBJ's Y-up convention to Rhino's Z-up coordinate system. The imported faces were consolidated into closed volumetric cells via `Topology.SelfMerge`. A semantic tag — ground, column, office, or core — was then attached to each cell along with a display colour, stored as a dictionary on a selector point placed inside the volume.

### Assembling the CellComplex

All tagged cells were merged into a single CellComplex. Wherever two spaces share a wall, that surface becomes an internal face that encodes the adjacency — each cell implicitly knows its spatial neighbours through shared boundaries.

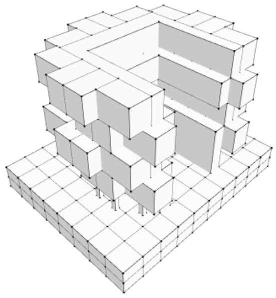


*Figure 1. CellComplex — merged volumetric model*

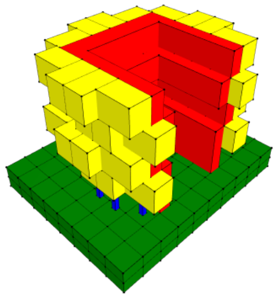


*Figure 2. CellComplex with type-based colouring (ground, columns, offices, core)*

## Propagating Labels

The selector dictionaries were transferred onto the unified cells using `Topology.TransferDictionariesBySelectors`, ensuring that every cell in the CellComplex retains its type label after the merge operation.

## Converting to a Graph

`Graph.ByTopology` converted the CellComplex into an attributed graph: each cell becomes a node and each shared internal face becomes an edge. Node types were one-hot encoded as five-dimensional feature vectors:

```
ground → [1, 0, 0, 0, 0]
column → [0, 1, 0, 0, 0]
office → [0, 0, 0, 1, 0]
core → [0, 0, 0, 0, 1]
```

The plinth category (index 2) was not used in this building. The feature matrix was exported to CSV as `feature_00` through `feature_04`, alongside node coordinates, edge lists, and the manually assigned graph label.

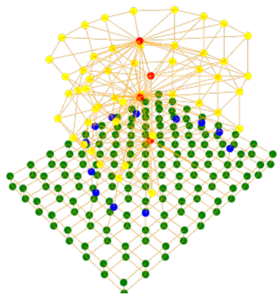


*Figure 3. Adjacency graph — nodes coloured by type, edges represent shared faces*

# Part 2 — Predicting the Label (13B)

## Training the Model

A GNN classifier was trained on the instructor-provided dataset (`dataset_graph_classification`) using PyTorch Geometric. The model performs graph-level classification into one of five building-ground relationship categories:

- **0 — Separation** — the building is raised on pilotis with no direct ground contact
- **1 — Separation with Plinth** — raised volume, but a solid base mediates the transition
- **2 — Adherence** — floor planes sit flush on the ground without an intermediary
- **3 — Adherence with Plinth** — grounded, but a podium extends and anchors the footprint
- **4 — Interlock** — the ground plane penetrates into the building volume, blurring the boundary

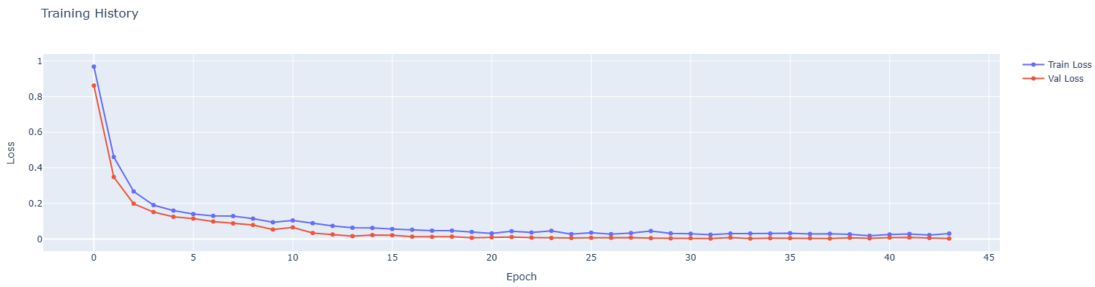



*Figure 4. Training and Validation Loss Curves*

## Prediction

The exported CSV dataset was loaded into a PyTorch Geometric data object and passed to the pre-trained model `bgr_model.pt` via the `Predict()` function. The output table compares the manually assigned label against the model's prediction and reports a confidence score.

| Actual Value | Predicted Value | Actual Label | Predicted Label | Confidence |
|---|---|---|---|---|
| 0 | 1 | Separation | Separation with Plinth | 1.0 |

*Table 1. Prediction results from bgr_model.pt*

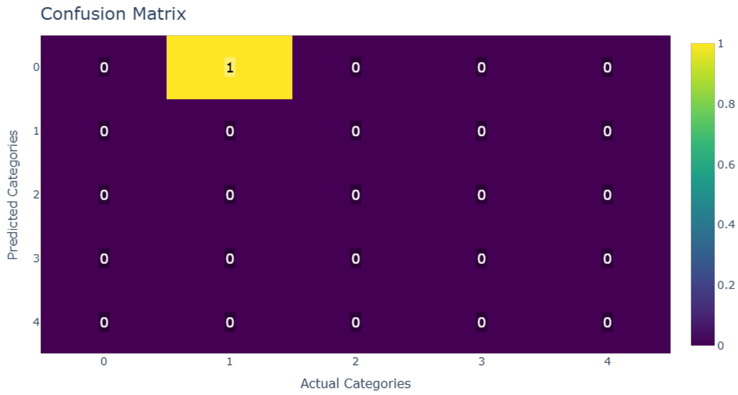



*Figure 5. Confusion Matrix*

# Reflection — Separation vs. Separation with Plinth

The manually assigned label was 0 (Separation), while the model predicted 1 (Separation with Plinth) with a confidence of 1.0. The discrepancy is meaningful and worth examining.

The building features a ground slab with columns lifting portions of the structure above it. The core has a single geometry whose base sits flush to the ground. The columns also sit flush on the ground slab. For the offices, only some first-level volumes rest on columns, while the upper levels stack on top of each other. Based on this reading — volumes touching the ground face-to-face but not overlapping — the assigned label of Separation seemed appropriate.

However, the GNN classified the graph as Separation with Plinth with full confidence. The model likely interprets the ground slab as a plinth — a continuous platform that mediates between the earth and the building above. From a topological perspective, the ground slab is a highly connected node adjacent to columns, core, and some office cells. This connectivity pattern resembles a plinth more than a simple separation layer, because the ground node acts as a shared base that multiple spatial types connect through rather than a passive surface beneath isolated pilotis.

This highlights a key insight about graph-based classification: the model reads spatial relationships through topology (adjacency), not geometry (shape, height, thickness). A thin ground slab and a thick plinth may produce identical graph structures, making them indistinguishable to the GNN. Richer node features — such as volume, height, or surface area — could help the model differentiate these cases.